Cyber Security Internship – Module 1

Internship Report

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CODEALPHA TECHNOLOGIES

Task 1 – Basic Network Sniffer

1. Objective

The objective of this task is to capture network traffic packets, analyze their source and destination IP addresses, and identify the type of protocol used (TCP/UDP).
This helps in understanding how data flows in a network and provides basic knowledge of network protocols.

2. Source Code

```
from scapy.all import *

def packet_callback(packet):
    print("\n--- New Packet ---")

    if packet.haslayer(IP):
        ip_layer = packet[IP]
        print(f"Source IP: {ip_layer.src}")
        print(f"Destination IP: {ip_layer.dst}")

    if packet.haslayer(TCP):
        print("Protocol: TCP")
    elif packet.haslayer(UDP):
        print("Protocol: UDP")
    else:
        print("Protocol: Other")

sniff(prn=packet_callback, count=10)
```

3. Procedure / Steps

1. Installed Python 3 and Scapy library.

2. Created a file named `sniffer.py` and pasted the above code.

3. Opened Command Prompt (CMD) or Terminal and navigated to the file location.

4. Ran the script: `python sniffer.py`.

5. Observed the captured packets displayed in the terminal.

4. Output

Image-1

```
E:\>python sniffer.py

--- New Packet ---
Source IP: 172.64.146.215
Destination IP: 192.168.1.15
Protocol: UDP
```

Image-2

```
--- New Packet ---
Source IP: 192.168.1.15
Destination IP: 172.64.146.215
Protocol: UDP
```

Image-3

```
--- New Packet ---
Source IP: 192.168.1.3
Destination IP: 224.0.0.251
Protocol: UDP
```

Image-4

```
--- New Packet ---
Source IP: 192.168.1.12
Destination IP: 239.255.255.250
Protocol: UDP
```

Image-5

```
--- New Packet ---
Source IP: 192.168.1.3
Destination IP: 224.0.0.251
Protocol: UDP

--- New Packet ---
Protocol: UDP
```

Image-6

```
--- New Packet ---
Source IP: 104.18.39.21
Destination IP: 192.168.1.15
Protocol: TCP
```

5. Explanation

The program uses Scapy library to capture live network packets. Each packet is analyzed to display its source and destination IP addresses along with the protocol type (TCP/UDP/Other).

This helps understand:

- How network data flows between devices

- The basic structure of network packets

- The difference between TCP and UDP protocols

6. Conclusion

Task 1 provided hands-on experience in capturing and analyzing network packets.

It helped enhance understanding of network protocols, packet structures, and practical Python scripting for cybersecurity tasks.